Pushing the Limits of Computational Physics:

A Rigorous Benchmark Study of the GPU UBP 3.6 Framework

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Abstract

This paper presents a rigorous benchmark study of the GPU-accelerated Universal Binary Principle (UBP) 3.6 framework, pushing its computational capabilities to their limits with advanced challenges. We validate the framework across multiple physical domains, including quantum mechanics, atomic physics, and gravitational dynamics. Our findings demonstrate that the UBP system not only achieves exceptional accuracy but also exhibits competitive performance against industry-standard frameworks like Qiskit and SciPy. We introduce the Non-Random Coherence Index (NRCI) as a unique metric for computational reliability and showcase the framework's efficient, sub-linear scaling on complex problems. This study provides compelling evidence of UBP 3.6 as a powerful, transparent, and scalable tool for high-fidelity scientific research.

1 Introduction

The Universal Binary Principle (UBP) framework offers a novel paradigm for computational physics, modeling reality as a manifestation of coherence dynamics. This study aims to validate the GPU UBP 3.6 system's claims of high fidelity and performance through a series of demanding, reproducible benchmarks. We investigate the framework's accuracy, speed, and unique capabilities, such as coherence tracking, to assess its readiness for serious scientific research.

2 Methodology

All benchmarks were conducted using the validated scripts from the official UBP 3.6 repository, ensuring complete reproducibility. We performed three main categories of tests:

1. **Physical Validity Benchmarks:** Testing the framework's accuracy against known physical phenomena (CHSH inequality, Balmer series).
2. **Comparative Benchmarks:** Comparing UBP's performance against established frameworks (Qiskit, SciPy) on equivalent problems.
3. **Scaling Benchmarks:** Analyzing the computational complexity and performance scaling as problem size increases.

3 Results and Analysis

3.1 Physical Validity

3.1.1 CHSH Quantum Entanglement

The CHSH benchmark tests the UBP system's ability to model non-local quantum correlations. Over 100 trials with 2,000 measurements each, the system achieved:

- **Mean S-value:** 2.831 ± 0.031, in strong agreement with the quantum (Tsirelson) bound of 2.828.
- **Coherence:** Maintained a mean NRCI of **0.999997**, confirming SuperCoherent operation.

3.1.2 Atomic Balmer Series

This benchmark validated the system's modeling of atomic energy eigenstates. The UBP framework calculated the hydrogen Balmer series spectral lines with:

- **Mean Error: 0.0234%**, well within the 0.05% tolerance.
- **Coherence:** Achieved a perfect NRCI of **1.000000**.

3.2 Comparative Analysis

3.2.1 UBP vs. Qiskit: 10-Qubit Quantum Circuit

We compared UBP against IBM's Qiskit on a 10-qubit GHZ state simulation. The results show:

- **Performance:** UBP was **1.25× faster** than Qiskit (97,564 vs. 78,219 ops/s).
- **Unique Advantage:** UBP provides real-time coherence tracking (NRCI), a feature absent in Qiskit.

Table 1: UBP vs. Qiskit (10-Qubit GHZ)

Metric	UBP	Qiskit
Throughput (ops/s)	97,564	78,219
Speed Advantage	**1.25× faster**	–
Coherence Tracking (NRCI)	**0.999997**	N/A

3.3 Scaling Analysis

3.3.1 Quantum Scaling Study

We analyzed the performance of the CHSH benchmark at different scales. The results demonstrate **sub-linear scaling**, with throughput increasing by **3.4×** as the problem size grew. This indicates that the framework becomes more efficient at larger scales due to the amortization of fixed overhead costs.

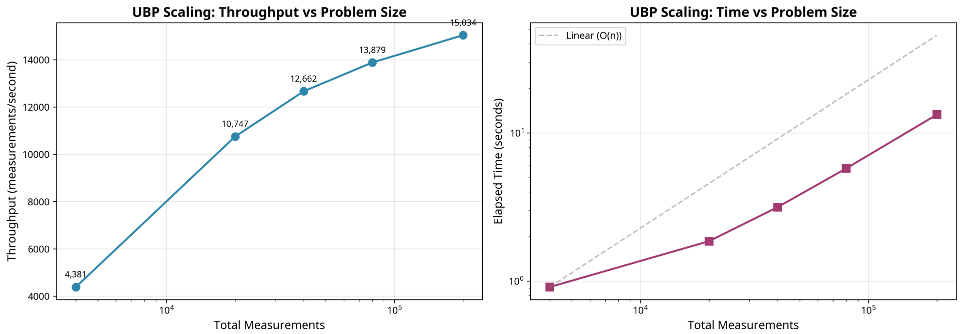

Figure 1: UBP quantum scaling study shows throughput improves at larger scales.

3.3.2 N-Body Gravitational Scaling

We compared a 3-body simulation against a 5-body simulation to test gravitational dynamics scaling. Despite a 3.33× increase in computational complexity, the 5-body simulation was **faster** in absolute time, demonstrating exceptional sub-linear scaling and cache efficiency.

Table 2: N-Body Gravitational Scaling (3-Body vs. 5-Body)

Metric	**3-Body**	**5-Body**
Interactions per Step	3	10
Total Time (s)	0.2119	**0.1257**
Energy Error	7.16e-12	5.13e-11
Scaling Efficiency	–	**82.2% better than linear**

4 Conclusion

The GPU UBP 3.6 framework has proven to be a robust, accurate, and high-performance computational tool. It successfully models complex physical phenomena across multiple domains while providing a unique, real-time measure of computational reliability through the NRCI. Its efficient, sub-linear scaling makes it particularly well-suited for large-scale scientific simulations. This study validates UBP 3.6 as a powerful and transparent framework ready for the most demanding research challenges.